

Devi Sri Ranga Prasad

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PROFESSIONAL SUMMARY

Backend Software Engineer with 1+ years architecting enterprise microservices at Merck KGaA, serving 5,000+ users. Specialized in high-performance Java applications with proven track record of 40% performance optimization, 70% automation efficiency gains, and delivering real-time data solutions at scale.

TECHNICAL SKILLS

Languages:	Java (8/11/17) • Python • JavaScript • C++ • C
Backend & Frameworks:	Spring Boot (2.x/3.x) • Spring WebFlux • Spring MVC • Microservices Architecture • RESTful APIs • Server-Sent Events (SSE)
AI/ML Frameworks:	LangChain • LangGraph • Model Context Protocol (MCP) • RAG Systems • NLP
Databases:	MySQL • MongoDB • Redis • InfluxDB • Elasticsearch
DevOps & Tools:	Docker • Git • CI/CD Pipelines • JIRA • Performance Tuning • System Integration
Integration & Monitoring:	Icinga2 • MuleSoft • IT4YOU • JPA • JDBC
Practices:	Agile/Scrum • Code Reviews • Unit Testing • Technical Documentation • TDD

PROFESSIONAL EXPERIENCE

Software Engineer

Merck KGaA, Darmstadt, Germany

Aug 2025 – Present

Full-time

- Developed scalable backend features using Spring Boot microservices, supporting 5,000+ enterprise users with 99.9% uptime
- Optimized system performance by implementing Redis caching layer that reduced database load by 40% and improved API response times from 2s to 500ms
- Built real-time monitoring solutions using Spring WebFlux and Server-Sent Events (SSE) for live dashboards, processing 50,000+ daily requests across 100+ screens
- Implemented automated incident response system integrating IT4YOU ticket generation, reducing manual intervention by 70% and improving response time by 60%
- Led troubleshooting and root cause analysis for production issues, maintaining comprehensive technical documentation for 25+ critical services
- Primarily worked on complex story points (8+), delivering high-impact features involving multiple system integrations and architectural decisions
- Mentored intern developers through code reviews, establishing coding standards and Spring Boot best practices across team of 15+ engineers

Software Engineering Intern

Merck KGaA, Darmstadt, Germany

Jan 2024 – Aug 2025

Internship

- Developed 25+ RESTful APIs using Java and Spring Boot, powering critical business functionality for 10,000+ users across 8 platforms
- Optimized backend services through advanced caching and query optimization, achieving 35% performance improvement in high-traffic endpoints
- Designed and implemented microservices components with MySQL, InfluxDB, and Elasticsearch, ensuring scalable data access patterns
- Collaborated with frontend teams on API contracts and integration testing, ensuring seamless user experience across web applications
- Contributed to architecture design discussions and sprint planning, providing technical insights for scalable system design decisions
- Achieved 95% code quality standards through comprehensive unit testing and rigorous peer code review participation
- Integrated monitoring solutions and performance metrics, enabling proactive identification of bottlenecks in production systems

KEY PROJECTS

Central Monitoring Platform (CMP) – [cmp.merckgroup.com](#)

Jan 2025 – Present

Tech Stack: Java • Spring Boot • Microservices • Redis • MySQL • InfluxDB • Elasticsearch • Icinga2 • MuleSoft

Overview: Enterprise-wide unified monitoring dashboard providing real-time application health, server metrics, and automated incident management for 200+ critical applications.

- Developed secure admin panel with role-based access control using Spring Security, managing 1,000+ stakeholder permissions
- Engineered Redis caching strategy reducing database queries by 40% and API latency from 2s to 500ms for 50,000+ daily requests
- Built real-time data pipeline using Spring WebFlux SSE for instant dashboard updates across 100+ monitoring screens with sub-second latency
- Developed RESTful services delivering environment-specific health metrics and status for 200+ applications using JPA and JDBC
- Built notification service consuming messages from Kafka topics and integrating with ticketing systems for centralized alert management
- Implemented scheduler services fetching data from multiple sources and publishing to Kafka topics for downstream processing

My Safety Agent – AI-Powered Voice Assistant

Jun 2024 – Present

Tech Stack: Python • LangChain • LLMs • Model Context Protocol (MCP) • RAG • NLP

Overview: Voice-enabled LLM-powered assistant using RAG architecture for hands-free access to chemical Safety Data Sheets (SDS) in laboratory environments.

- Migrated legacy system to Model Context Protocol (MCP) architecture, improving modularity by 60% and reducing deployment time by 45%
- Designed RAG pipeline for intelligent document retrieval and contextualized responses from 1,000+ SDS documents
- Ensured 100% backward compatibility during modernization while serving 200+ laboratory users across multiple sites
- Implemented natural language processing for voice command interpretation with 90%+ accuracy rate

My Coding Assistant – AI Development Tool

Mar 2024 – Sep 2024

Tech Stack: Python • LangChain • NLP • Machine Learning • Code Analysis Tools

Overview: AI-powered coding assistant for automated code generation, intelligent debugging, and quality analysis across enterprise codebases.

- Built intelligent code generation and debugging system reducing development time by 30% for team of 10 developers
- Implemented ML-based bug pattern detection achieving 85% accuracy across 50+ production codebases
- Integrated LangChain framework for advanced NLP enabling natural language to code translation capabilities
- Automated code quality analysis providing actionable recommendations, reducing code review time by 25%
- Created automated testing suggestion engine improving test coverage from 60% to 85% across projects

EDUCATION

Bachelor of Technology in Computer Science and Engineering <i>Lovely Professional University, Punjab, India</i>	2025 <i>CGPA: 7.7/10.0</i>
Relevant Coursework: Data Structures & Algorithms • Database Management Systems • Software Engineering • Object-Oriented Programming • System Design • Computer Networks	
Higher Secondary School Certificate <i>Narayana Junior College, Vijayawada, India</i>	2020 <i>Percentage: 96.5%</i>
Secondary School Certificate <i>Narayana School, Guntur, India</i>	2018 <i>CGPA: 10.0/10.0</i>

CERTIFICATIONS

- **Generative AI Certification** – FreeCodeCamp (2024)
- **Java Spring Boot & Spring Reactive Certification** – FreeCodeCamp (2024)

ACHIEVEMENTS & RECOGNITION

- **Competitive Programming Excellence:** Top 5% globally – LeetCode (1600+) • Codeforces (1300+) • CodeChef (1400+) • GeeksforGeeks (1700+)
- Promoted from Intern to Full-time Software Engineer within 12 months based on exceptional performance and technical contributions
- Recognized for delivering mission-critical features ahead of sprint deadlines while maintaining 99.9% production uptime